

Research Note

Perceived Evaluative Threat and State Anxiety During Exercise in Women with Social Physique Anxiety

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Self-presentation represents the processes by which individuals monitor and attempt to control how others perceive them (Schlenker & Leary, 1982). The impressions others form have implications for virtually all facets of our social lives. Given the importance of physical attractiveness in modern society, it is not surprising that self-presentational processes are particularly relevant in regard to physical attractiveness (Leary & Kowalski, 1990). Many individuals exercise for self-presentational reasons (Leary, 1992). Although women frequently cite self-presentational motives for engaging in regular physical activity (Hausenblas & Martin, 2000), concerns regarding the appearance of one's physique may serve as a barrier to exercise participation for some women. Social physique anxiety (SPA), a subtype of social anxiety defined as the disposition to experience apprehension in response to the presence or prospect of negative physique evaluation (Hart, Leary, & Rejeski, 1989), has been suggested to impede regular exercise participation. In this regard, results from research examining the relationship between SPA and exercise behavior reveals that women with SPA report preferences for private exercise environments (Spink, 1992) and group fitness settings in which less revealing attire is worn (Crawford & Eklund, 1994). Furthermore, SPA consistently demonstrates an inverse relationship with exercise participation (Lantz, Hardy, & Ainsworth, 1997; Treasure, Lox, & Lawton, 1998).

The presence or prospect of evaluative threat can provoke aversive affective responses among individuals with heightened self-presentational concerns (Leary & Kowalski, 1990; Schlenker & Leary, 1982). Thus, consistent with the tenets of self-presentational theory, those with high trait SPA may be predisposed to experience unfavorable changes in psychological states (i.e. negative affect, state anxiety) during exposure to exercise environments that provoke perceptions of negative physique evaluation. Several recent investigations have examined this

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relationship. For example, Focht and Hausenblas (2003) found that inactive women with high SPA reported increased state anxiety while exercising in a naturalistic setting which elicited the greatest perceptions of evaluative threat. Katula and colleagues (Katula, McAuley, Mihalko, & Bane, 1998) observed that a sample of moderately active women reported lower self-efficacy while exercising in front of a mirror. Additionally, SPA was found to be a significant independent predictor of self-efficacy in the mirrored exercise environment. In another recent investigation, sedentary women reported negative feeling state responses following exercise performed in front of a mirror (Martin-Ginis, Jung, & Gauvin, 2003). It should be noted that multi-level modeling analysis revealed that negative feeling states emerged following exercise in the mirrored environment irrespective of one's predisposition towards body image concerns. Nevertheless, taken together, these findings suggest that self-presentational aspects of fitness settings (i.e., presence of other exercisers and mirrors) influence state anxiety, self-efficacy, and feeling states during and following exercise and this effect may be moderated by one's trait level of SPA. Given the prospect of bodily evaluation present in most public fitness facilities and the potentially deleterious impact that psychological distress resulting from perceived evaluative threat may have on long-term exercise participation (Bain, Wilson, & Chaikind, 1989), further exploration of factors that influence state anxiety during exercise in women with SPA is warranted.

Concerns involving limitations with nomothetic conceptions of the exercise-affect relationship have recently emerged (VanLanduyt, Ekkekakis, Hall, & Petruzzello, 2000). VanLanduyt et al. state: "Acknowledging the role of individual differences when studying the effects of different exercise doses on affect is very important" (p. 210). Thus, while recent evidence suggests that self-presentational aspects of the environment may systematically influence psychological responses to exercise (Focht & Hausenblas, 2003; Katula et al., 1998; Martin-Ginis et al., 2003), it is important to recognize that individual differences may mediate this effect. For example, an environment providing the prospect of bodily evaluation may have little impact upon affective responses during exercise if one does not perceive that the potential for physique evaluation is sufficiently threatening. Therefore, while we have demonstrated that women with high SPA report increased state anxiety while exercising in a public fitness facility (Focht & Hausenblas, 2003), it is likely that individual differences in the perceptions of evaluative threat elicited by the environment influenced the state anxiety responses observed during exercise. Nevertheless, the extent to which such individual differences may impact the pattern of state anxiety responses observed during exercise has yet to be examined.

In regard to the relevance of individual difference in the exercise-affect relationship, mounting evidence suggests that baseline values are important determinants of the affective responses to exercise. Notably, participants most likely to demonstrate the affective benefits of acute exercise were those who reported less positive or more negative psychological states prior to exposure to the exercise stimulus (Focht, 2002; Rejeski, Gauvin, Hobson, & Norris, 1995). Given these findings, it is possible that individual differences in perceived evaluative threat and baseline scores may systematically influence state anxiety during exercise in women with SPA. Consequently, in the present investigation we have conducted an ancillary analysis of our prior research (Focht & Hausenblas, 2003) to examine this possibility. Therefore, the primary objective of this investigation is to examine the influence of environmentally-induced perceptions of evaluative threat upon the state anxiety responses during exercise among women with high SPA. The secondary purpose of this investigation is to examine the if differences in baseline scores influence state anxiety responses during exercise. It is hypothesized that women reporting the highest perceived evaluative threat during exercise in the naturalistic environment will demonstrate the greatest increase in state anxiety. Additionally, it is hypothesized that the most favorable changes in state anxiety during exercise will be exhibited by those with the highest baseline values.

METHOD

Participants

Thirty female students (M age = 20.20 years; SD = 1.65; 77% Caucasian; 10% African American; 10% Hispanic American; 3% Asian American) enrolled at a university in the south-eastern United States were recruited to participate in this study. In order to obtain a sample of high SPA women with low physical activity levels, only volunteers reporting two or fewer bouts of moderate or vigorous exercise during a typical week and scoring 36 or above on the Social Physique Anxiety Scale (Martin, Rejeski, Leary, McAuley, & Bane, 1997) were eligible to participate. All volunteers completed a medical history questionnaire and an informed consent document prior to participating in the study.

Measures

Social Physique Anxiety Scale

The nine-item Social Physique Anxiety Scale (SPAS) was used to assess participants' trait disposition to experienced anxiety as a function of other's evaluation of their physique (Martin et al., 1997). Participants indicated the degree that each item was characteristic of them on a 5-point Likert scale. SPAS scores range from 9 (low) to 45 (high). Prior research suggests that the 9-item SPAS demonstrates acceptable convergent and divergent validity and adequate reliability (Hausenblas & Martin, 2000; Martin et al., 1997). Consistent with these findings, the SPAS exhibited good internal consistency in this study (α = .86).

Leisure-Time Exercise Questionnaire

The Leisure-Time Exercise Questionnaire (LTEQ) measures the self-reported frequency of strenuous, moderate, and mild bouts of exercise during a typical week (Godin & Shephard, 1985). The LTEQ has demonstrated adequate reliability and validity in previous research (Jacobs, Ainsworth, Hartman, & Leon, 1993).

State Anxiety Inventory

State anxiety was measured using the 20-item version of the State Anxiety Inventory (SAI, Spielberger, Gorsuch, Lushene, Vagg, & Jacobs, 1983). Participants responded to each item on a 4-point Likert scale ranging from 1 (*not at all*) to 4 (*very much so*), with higher scores indicating higher state anxiety. The SAI demonstrated adequate internal consistency in the present investigation with alpha values ranging from .83 to .91.

Perceived Evaluative Threat

Participants were asked to rate their level of perceived evaluative threat within each environment. Consistent with procedures used in previous research (Hart et al., 1989), participants were specifically asked to rate how threatening this exercise environment was in terms of having your physique evaluated. Participants provided ratings using a 5-point Likert scale ranging from 0 (*not at all*) to 4 (*extremely*).

Procedures

A more comprehensive description of the procedures is provided elsewhere (Focht & Hausenblas, 2003). However, we have provided a brief overview of this information. Potential participants completed the SPAS and LTEQ during lab and lecture classes. Thirty women meeting the eligibility criteria were subsequently contacted and volunteered to participate.

Fifteen participants were randomly assigned to either a self-selected or imposed intensity group (70–80% of HRMax).

Each participant completed two 20 min bouts of stationary cycling. One bout was completed in a naturalistic exercise environment while the other bout was performed in a laboratory exercise environment. The naturalistic exercise condition was conducted at a university fitness facility. Participants exercised on a stationary cycle located in front of a full length mirror which was situated in a manner that allowed the participants to view themselves. An average of 36 ($SD = 12.27$) other recreational exercisers (14 Men, $SD = 5.69$; 22 Women, $SD = 7.89$) were present in the fitness center during the naturalistic condition. The laboratory exercise condition was performed in a private laboratory setting without mirrors or the presence of other exercisers. To maintain the private nature of the environment, interaction between the participant and investigator was limited to the assessment procedures. At all other times during the laboratory condition, the experimenter remained out of view and refrained from social interaction with the participant. Exercise conditions were presented in a counterbalanced fashion and participants completed the exercise sessions on separate days. Women with SPA may use protective self-presentational strategies such as the wearing of baggy clothing to conceal the appearance of their physique (Leary, 1992). To control for this potentially confounding influence upon perceptions of evaluative threat, exercise attire was standardized during each condition as participants were required to wear a t-shirt and a pair of shorts to each session.

Upon arrival at the laboratory, participants completed the baseline state anxiety assessment and were then fitted with the Polar Vantage heart rate monitor. In an attempt to minimize the influence of demand characteristics or expectancy effects on the findings, participants were only informed which condition they would be performing after they completed the baseline measurement. State anxiety was assessed prior to, during (15th min of activity) and following each condition. However, for the purposes of this investigation, analyses were limited to change in state anxiety observed during exercise. Perceived evaluative threat was assessed after the completion of the exercise session.

RESULTS

As reported elsewhere (Focht & Hausenblas, 2003), the naturalistic exercise environment produced significantly higher ($p < .001$) ratings of perceived evaluative threat than the laboratory exercise environment. Thus, the experimental manipulation was successful in creating environments that differed in the perceived presence and prospect of self-presentational physique threat. Results of this prior analysis also revealed that state anxiety during exercise did not differ between the imposed and self-selected intensity groups. Consequently, the imposed and self-selected intensity data was pooled within each exercise environment for the ancillary analysis reported in this investigation. To examine the role of evaluative threat upon the state anxiety responses to exercise, ratings of perceived evaluative threat for each environment were dichotomized into high and low evaluative threat groups via a median split. State anxiety responses during exercise were then analyzed using a 2 (Evaluative threat: high and low) $\times$ 2 (Environment: naturalistic and laboratory) ANCOVA with the baseline value as the covariate. The dependent measure in the ANCOVA was change in state anxiety calculated as the SAI responses obtained during exercise minus the SAI baseline value. Tukey post-hoc analysis was performed to probe for significant differences among the mean changes. ANCOVA analysis of change in SAI during exercise yielded a significant Baseline $\times$ Evaluative Threat $\times$ Environment interaction, $F(4, 52) = 5.90$, $p < .001$. None of the remaining main effects or two-way interaction effects reached statistical significance ($p > .35$). Post hoc analysis revealed that

Table 1
Mean (*SE*) Changes in State Anxiety During Exercise in the Naturalistic and Laboratory Environment

	Naturalistic	<i>ES</i>	Laboratory	<i>ES</i>
High evaluative threat	5.87 (1.55)*	—	1.88 (1.71)	-.53
Low evaluative threat	1.74 (1.67)	-.61	-0.18 (1.49)	-.75

* = $p < .001$.

Note: *ES* calculated by taking the mean difference from the high evaluative threat group and dividing by the pooled standard deviation.

the change in state anxiety reported during exercise in the naturalistic environment by women in the high perceived evaluative threat group was responsible for the significant interaction effect. Specifically, after controlling for baseline values, the increase in state anxiety reported by the high perceived evaluative threat group while exercising in the naturalistic environment was significantly different from any of the other mean changes observed during exercise with effect sizes ranging from $-.53$ to $-.75$ (see Table 1). However, inspection of the predicted regression lines depicted in Figure 1 better illustrates the nature of the significant 3-way interaction effect. Specifically, the graph demonstrates that women reporting the lowest baseline state anxiety in the high perceived evaluative threat group demonstrated the largest increase in state anxiety during exercise in the naturalistic environment.

DISCUSSION

As illustrated in Figure 1, women reporting the lowest baseline anxiety and highest perceived evaluative threat exhibited the greatest increase in state anxiety while exercising in

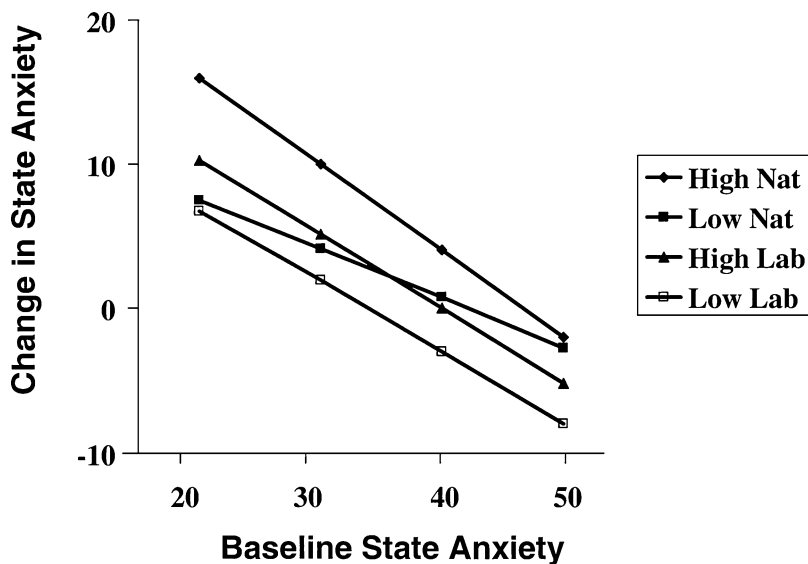


Figure 1. Combined effect of baseline, perceived evaluative threat, and environment on changes in state anxiety during exercise.

the naturalistic environment. Thus, perceptions of evaluative threat present in the exercise environment interact with baseline values to influence state anxiety during exercise in women with high SPA. The findings are consistent with the results of previous research demonstrating that self-presentational aspects of the exercise environment can negatively impact the psychological responses to exercise (Focht & Hausenblas, 2003; Katula et al., 1998; Martin et al., 2003) and baseline values influence affective responses during and following exercise (Focht, 2002; Rejeski et al., 1995). Furthermore, the present findings also support the contention that meaningful changes in affective responses can be masked in aggregate analyses (VanLanduyt et al., 2000). Therefore, individual differences in baseline state anxiety and perceptions of evaluative threat during exercise should be considered salient determinants of the anxiolytic effect of exercise among women with high SPA in future research.

It has been contended that increases in state anxiety frequently documented during exercise in past research are an artifact due to the effect of physical exertion upon inventory items tapping physiological arousal or activation (Ekkekakis, Hall, & Petruzzello, 1999). Although a comprehensive evaluation of the SAI during exercise is beyond the scope of this study, it should be noted that the significant increase in state anxiety observed during exercise in the naturalistic environment was comprised of elevations in SAI items reflecting both anxiety and activation. Expanded discussion of this result and its implications for the use of the SAI during exercise is provided elsewhere (Focht & Hausenblas, 2003).

The increase in state anxiety during exercise in the naturalistic environment suggests that aspects common to many public exercise environments (i.e., mirrors and the presence of other exercisers) can heighten perceptions of evaluative threat and psychological distress during exercise. When considering this finding in conjunction with the growing prevalence of body image concerns in modern society, the present findings have important implications for the promotion of physical activity among women. One implication is the psychological consequences associated with the placement of exercise equipment and mirrors must be carefully considered. Modifying public fitness facilities to include sections without mirrors or exercise areas that provide more privacy may be beneficial for women with high SPA, particularly those in the process of initiating exercise participation. Many fitness centers now provide women-only exercise rooms and recent evidence suggests that women who exercise in these areas report greater body dissatisfaction than women who exercise in the common areas of the same fitness center (Yin, 2001). However, sections of fitness centers or exercise classes designed exclusively for women may also cultivate increased physique comparison among female exercisers (Markula, 1995). Accordingly, additional research is necessary to determine if modifying potentially threatening self-presentational aspects of fitness facilities significantly impacts affect during exercise or exercise participation among women with body image concerns.

Due to the diverse aspects inherent to the naturalistic exercise environment (presence of other exercisers and mirrors), the mechanisms responsible for the increased perceived evaluative threat and state anxiety cannot be determined at the present time and this represents a primary limitation of the present investigation. Nevertheless, there are some plausible explanations for the present findings. It is possible that exercising in front of mirrors increases self-awareness and elicits a self-evaluation process that produces negative affective responses in women with SPA. Conversely, it is also reasonable to suggest that the increase in state anxiety resulted from the high SPA women's perception of unfavorable physique comparisons between themselves and other female exercisers in the naturalistic exercise facility. Although each of these explanations remains tenable, future research is needed to disentangle the influence of increased self-awareness and negative perceptions of physique comparisons upon affective responses during exercise.

The present study has other limitations that should be acknowledged. First, it is likely that the single-item measure of perceived evaluative threat used in this investigation provides a blunt index of this construct. Accordingly, we believe that the development of a more comprehensive measure of perceived evaluative threat would be advantageous for future research in this area. In this regard, it has been speculated that SPA, while conceptualized and currently measured as a stable psychological trait, may demonstrate labile properties within socially evaluative conditions (Focht & Hausenblas, 2001; Katula et al., 1998). Consequently, progress in the identification and development of a measure that captures potential state-like fluctuations in SPA could enhance current understanding of the SPA construct and its impact upon exercise participation. Dichotomizing the sample into groups on the basis of perceptions of evaluative threat reduces the statistical power of this ancillary analysis. Thus, replication of these findings in a larger sample is necessary to substantiate the veracity of the conclusions presented herein. Finally, the present sample was restricted to low active college age women. Hence, it would be inappropriate to generalize these findings to more physically active or women of different ages.

In summary, the results of this study suggest that changes in state anxiety during exercise are influenced by individual differences in perceptions of evaluative threat and baseline anxiety level. Affective responses to acute exercise and body image concerns may serve as important determinants of exercise participation for women. Consequently, future inquiry delineating the relationship between self-presentational concerns and exercise behavior may contribute to the development of more effective interventions for the promotion of physical activity in women.

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